

STRANGLER-IPU: An Ingress-Aware Evolutionary Processing Architecture for High-Velocity Compute-Memory Interconnects

Research Specification & Architectural Technical Report

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Chapter 1

Introduction & Theoretical Formulation

1.1 Background: The Von Neumann Memory Chasm

Modern computing architectures are fundamentally constrained by the physical and thermodynamic energy penalties associated with data movement across the Von Neumann chasm. In traditional accelerator systems, fetching an operand from off-chip dynamic random-access memory (DRAM) across physical bus traces incurs two to three orders of magnitude higher energy dissipation than the actual arithmetic multiply-accumulate (MAC) operation.

$$\frac{E_{\text{DRAM_FETCH(32-bit)}}}{E_{\text{ALU_FMA(32-bit)}}} \geq 100\times$$

While proprietary accelerator vendors (e.g., NVIDIA Hopper and Blackwell architectures) scale high-bandwidth memory (HBM) interfaces via advanced 2.5D/3D packaging technologies (e.g., TSMC CoWoS), such brute-force bandwidth scaling does not eliminate the fundamental thermodynamic dissipation or queue latency inherent in moving semantically reducible data across the host memory hierarchy.

1.2 Problem Statement: High-Velocity Ingress Mismatch

In emerging high-throughput workloads—ranging from 6G Sub-THz massive MIMO baseband front-ends to long-context Transformer key-value (KV) cache retrieval—the rate of incoming data generates extreme pressure at the host interconnect boundary. When high-velocity streams possess low semantic density (i.e., where only a fraction of raw bits represent actionable state), delivering raw bytes directly to host DRAM before execution creates severe bus contention and wasted memory bandwidth.

1.3 Research Objectives

This research formulates and evaluates an alternative architectural paradigm governed by the following core questions:

1. **Traffic Reducibility:** Can programmable semantic processing directly at the ingress boundary reduce interconnect pressure more efficiently than scaling raw bus bandwidth?
2. **Break-Even Frontier:** Under what ingress rates (R_{in}) and semantic density ratios (ρ) does boundary processing strictly dominate conventional host-side architectures?

3. **Evolutionary Coexistence (The STRANGLER Model):** Can an Ingestion Processing Unit (IPU) be introduced transparently without demanding application-level rewrites or obsolete standard CXL/PCIe protocols?

Chapter 2

System Architecture & Ingestion Processing Unit (IPU)

2.1 The STRANGLER Evolutionary Paradigm

Clean-slate architectures typically fail commercial deployment when they require complete replacement of established software ecosystems (e.g., CUDA, PyTorch) or physical motherboard standards. Analogous to the multi-decade coexistence of cellular generations (2G, 3G, 4G, and 5G), the **STRANGLER** model introduces an evolutionary, bump-in-the-wire ingress processor that attaches directly to existing interconnect standards:

- **Phase 1 (Transparent Compatibility):** The IPU acts as an enhanced memory controller or smart NIC, executing dynamic request coalescing and burst packing without requiring host driver modifications.
- **Phase 2 (Opportunistic Stream Processing):** High-volume streaming reductions, baseband channel estimations, and attention pre-filters are intercepted and computed in-flight before entering host memory.
- **Phase 3 (Architectural Dominance):** As boundary reduction handles the vast majority of raw data movement, monolithic host accelerators transition into lightweight executive coordinators.

2.2 IPU Subsystem Topology

The Ingestion Processing Unit consists of three decoupled hardware stages:

1. **Line-Rate Ingress Buffer:** Absorbs multi-gigabit bursts directly from high-speed PHY interfaces (Sub-THz ADC, Ethernet, SerDes) to eliminate host interrupt storms.
2. **In-Flight Stream Transform Engine:** Executes streaming arithmetic (matrix reductions, FFT pipelines, vector dot products) locally next to the ingress buffer.
3. **Interconnect Adapter:** Emits reduced semantic payloads formatted as standard, cacheline-aligned bursts over PCIe 5.0/6.0 or CXL 3.0/4.0 channels.

Chapter 3

Empirical Evaluation & Break-Even Parameter Sweeps

3.1 Experimental Methodology (EXP-001 Protocol)

To evaluate the boundary at which the IPU outperforms conventional host memory architectures, a multi-dimensional parameter sweep was executed modeling varying ingress rates $R_{\text{in}} \in [25, 500]$ GB/s and semantic compression ratios $\rho \in [0.001, 0.90]$ across three generations of host interconnects:

1. **PCIe 5.0 x16:** 64 GB/s
2. **CXL 3.0 / PCIe 6.0 x16:** 128 GB/s
3. **Next-Gen CXL 4.0:** 256 GB/s

3.2 Parameter Sweep Telemetry

Table 3.1 summarizes empirical latency and bus traffic metrics comparing the Conventional Host Pipeline against the STRANGLER-IPU Architecture across a representative CXL 3.0 interconnect topology (128 GB/s).

Table 3.1: EXP-001 Empirical Sweep Telemetry under CXL 3.0 Interconnect (128 GB/s)

R_{in} (GB/s)	ρ	Conv (μs)	IPU (μs)	Speedup	Traffic Saved	Winner
25	0.010	602.0	402.3	1.50×	99.0%	IPU
25	0.100	602.0	418.3	1.44×	90.0%	IPU
25	0.500	602.0	489.6	1.23×	50.0%	IPU
50	0.010	402.0	202.3	1.99×	99.0%	IPU
50	0.100	402.0	218.3	1.84×	90.0%	IPU
100	0.010	302.0	102.3	2.95×	99.0%	IPU
100	0.100	302.0	118.3	2.55×	90.0%	IPU
200	0.010	280.1	52.3	5.36×	99.0%	IPU
500	0.010	280.1	22.3	12.57×	99.0%	IPU

3.3 Analysis: The Break-Even Frontier

The results demonstrate that whenever the semantic reduction ratio satisfies $\rho \leq 0.50$ (i.e., at least half the incoming raw data represents redundant transmission entropy), the IPU achieves

strictly superior end-to-end latency and interconnect energy efficiency. Furthermore, when R_{in} exceeds host bus capacity, conventional host queues experience catastrophic latency degradation, whereas the IPU maintains bounded processing time.

Bibliography